

Kanglin Xu

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Education

University of Illinois Chicago, Chicago, IL Ph.D. in Computer Science	<i>Aug 2025 – Expected May 2030</i> GPA: 4.0
Texas Tech University, Lubbock, TX B.S. in Computer Science; Minor in Mathematics	<i>Aug 2021 – Dec 2024</i> GPA: 4.0
Lake Superior College, Duluth, MN General Study	<i>Aug 2020 – May 2021</i> GPA: 3.75
Hermantown High School, Duluth, MN General Study	<i>Aug 2017 – May 2020</i> GPA: N/A

Research Profile

My research lies at the intersection of machine learning and high-performance computing (HPC), with an emphasis on runtime prediction, reinforcement learning for system optimization, and AI-driven energy efficiency. I build predictive models from scheduler and system logs and develop uncertainty-aware buffering strategies to reduce underestimation and improve scheduling reliability. I am also exploring energy-aware scheduling and decision policies that improve throughput and reduce energy consumption in large-scale HPC systems.

Research Interests: HPC scheduling, batch job runtime prediction, energy-aware optimization, reinforcement learning, workload characterization, trustworthy ML for systems.

Skills

- **Programming:** Python, C/C++, Bash, Git
- **ML/DS:** TensorFlow / PyTorch, scikit-learn, XGBoost, Pandas/NumPy
- **HPC:** Linux, job schedulers (e.g., SLURM/PBS), profiling/monitoring workflows, cluster experimentation
- **Security:** vulnerability assessment support, policy/risk review, audit-oriented documentation

Professional Experience

Research Assistant, SPEAR Lab, University of Illinois Chicago (Chicago, IL)

Aug 2025 – Present

- Develop and implement reinforcement learning algorithms for HPC system optimization.
- Design and run large-scale experiments on HPC clusters to evaluate scalability and efficiency of RL-based approaches versus traditional heuristics.
- Collaborate on performance analysis by integrating learning-based policies with HPC workflows.

Information Security Technician, CoNetrix Security / CoNetrix (Lubbock, TX)

Jan 2023 – May 2025

- Conduct internal/external vulnerability assessment support using commercial and proprietary tools in coordination with audit and penetration testing groups.

- Review policies, procedures, risk assessments, and resiliency plans with information security consultants.
- Contribute to team-based solutions addressing evolving security requirements.

Research Assistant, Wu Research Group, Texas Tech University (Lubbock, TX)

Sep 2022 – May 2025

- Apply machine learning methods to analyze large medical datasets and support diagnosis-related research.
- Design and develop neural network models for clinical decision support tasks.
- Compare algorithms through systematic evaluation and visualization of results.

Teaching Assistant, Department of Computer Science, Texas Tech University (Lubbock, TX)

Sep 2024 – Dec 2024

- Support course delivery through grading, office hours, and student feedback.
- Assist with coursework evaluation (assignments, quizzes, exams) and learning support.

Projects

HPC Batch Job Runtime Prediction (Chicago, IL)

Runtime Modeling and Scheduling Support for HPC

Aug 2025 – Present

- Build machine learning models to predict batch job wall-clock runtime from scheduler logs (e.g., requested time, job size, queue/partition, and user/job metadata).
- Engineer workload- and user-aware features (e.g., rolling history of runtimes, recent underestimation patterns, and submission behavior) and compare global vs. personalized buffer strategies.
- Evaluate performance using RMSE, R^2 , underestimation rate, and operational implications for scheduling efficiency under typical queueing/backfilling policies.

Multiagent Deep Deterministic Policy Gradient Based Card Game (Lubbock, TX)

Multiagent Reinforcement Learning Project

Jan 2024 – May 2024

- Design and train AI agents to play a card game using agent–environment reinforcement learning.
- Implement a multiagent deep deterministic policy gradient (MADDPG-inspired) framework for decision-making.
- Visualize real-time game states and actions for both trained agents and human players.

Publications

1. B. Rezaei et al., “Magnetic Nanoparticles: A Review on Synthesis, Characterization, Functionalization, and Biomedical Applications,” *Small*, vol. 20, no. 5, p. 2304848, Feb. 2024. doi: 10.1002/smll.202304848.
2. S. Mostufa et al., “Giant Magnetoresistance Based Biosensors for Cancer Screening and Detection,” *ACS Applied Bio Materials*, vol. 6, no. 10, pp. 4042–4059, Oct. 2023. doi: 10.1021/ac-sabm.3c00592.
3. S. Mostufa, P. Yari, B. Rezaei, **K. Xu**, and K. Wu, “Flexible Magnetic Field Nanosensors for Wearable Electronics: A Review,” *ACS Applied Nano Materials*, vol. 6, no. 15, pp. 13732–13765, Aug. 2023. doi: 10.1021/acsanm.3c01936.

4. S. Mostufa et al., “Metamaterial as perfect absorber for high sensitivity refractive index based biosensing applications at infrared frequencies,” *Journal of Physics D: Applied Physics*, vol. 56, no. 44, p. 445104, Nov. 2023. doi: 10.1088/1361-6463/aceb6f.
5. P. Yari et al., “Static and dynamic magnetization models of magnetic nanoparticles: an appraisal,” *Physica Scripta*, vol. 98, no. 8, p. 082002, Aug. 2023. doi: 10.1088/1402-4896/ace8d1.

Presentations

- Undergraduate Research Conference, Texas Tech University (2024): *Image Based Deep Learning in ECG Signal*.
- Undergraduate Research Conference, Texas Tech University (2023): *Exploring the Effectiveness of Machine Learning Algorithms in Analyzing Cardiac Rhythms*.
- Computer Science Department External Advisory Board Meeting (2024): *Multiagent Deep Deterministic Policy Gradient Based Card Game*.

References

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